

SOCRATES field guide

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1. Basic resources

SOCRATES is a flexible real-gas radiative transfer code built and maintained at the Met Office in the UK. It is briefly described by the User Guide and Technical Guide available on the [Met Office Science Repository Service](#) (requires an account provided by the Met Office).

The GCM ROCKE-3D provides another guide available [here](#) along with a [repository](#) of spectral files. Different versions of SOCRATES are available without a Met Office account [here](#).

Spectral files can be easily generated using [this](#) public Github repository.

2. Setup in Bash

The compilers should first be adapted to the ones available on a given server, in make/Mk_cmd. In there, also adapt the path to netCDF and the openMP compiler flag.

Go into sbin/ and change the header “/bin/ksh” to “/bin/bash” in all files in sbin/ as well as ./examples/prsc/run_me. As of SOCRATES v.2511, this is done by default.

After loading the netCDF and HDF5 modules, compile SOCRATES with ./build_code

Activate the SOCRATES environment with “./set_rad_env”

Test that everything works as it should by executing ./quick_tests in examples/

3. Building inputs

The netCDF version of SOCRATES needs to be provided with input arrays contained in netCDF files, for example using the Python package nctoolkit from NCAR. The 2D quantities are surface albedo, surface temperature, surface pressure, solar zenith angle, and the irradiance at the top of the atmosphere. The 3D quantities are the temperature at the midpoints and edges of layers, the pressure, and the specific humidity.

Each quantity has a specific identifier, header, and file suffix visible in bin/input_head_pcf.f90.

4. Spectral file generation

Spectral files in SOCRATES are text files that contain all the necessary information about the features of the atmosphere that depend on wavelength. This includes the number of

spectral bands, the number of species in each band, their opacity coefficients, Rayleigh scattering coefficients, thermal source functions, stellar spectra, continua, aerosols, droplets, and ice crystals.

Once “./set_rad_env” is set, the manual pages become accessible. For example, “man Ccorr_k” will display the available options for computing correlated-k tables to add in a spectral file, and “man Cl_run_cdf” will show the different configurations available to run the netCDF version of the model.

The utility prep_spec is run first. It creates the skeleton file, which only includes the bands, absorbers, and continua. It is also used to add or replace data blocks in the spectral file.

Block 5 contains the k-terms for gaseous absorption and requires computation of k-tables with the utility corr_k (which uses interactive prompts), or Ccorr_k (the executable which accepts flags in lieu of prompts).

For a typical gas, it is run like so:

```
Ccorr_k -F ${GAS_DATA_DIR}/{PT_FILE} \  
-R 1 N_BANDS -l ID_GAS ${COL_MASS_K_GAS} -b 5.0e-4 \  
-s $skelfile +p -lk \  
-o $sp_dir/gas_lwf_l -m $sp_dir/gas_lwf_lm \  
-L $GAS_LBL_LWF \  
-sm $sp_dir/gas_lwf_l_map.nc \  
> $sp_dir/gas_lwf_log
```

PT_FILE is the pressure and temperature table. Each k-term will be computed for every value defined in this table.

ID_GAS is the gas identifier. It can be found in /bin/gas_list_pcf.f90.

COL_MASS_K_GAS is the maximum absorptive pathlength (kg/m²) of the gas.

Other inputs are described in the Ccorr_k manual page.

The continuum data is also computed using Ccorr_k, whose outputs are used by prep_spec to build Block 18 with the flag -CIA. Block 8 is the old continuum block which used the flag -ct. It involved fitting functions to approximate the continuum absorption. It was replaced by Block 18 and the flag -CIA, which is based on calculating k-tables.

Some gases or continua require specific flags. For example, the MT_CKD continuum of water vapor requires its bespoke -e flag.

The outputs in the example above are gas_lwf_l, which contains the correlated-k terms for each pressure and temperature, the monitoring file gas_lwf_lm, and gas_lwf_l_map.nc

which contains the mapping from wavenumber-space to g-space and the corresponding k-term weights.

Adding a specific gas to Block 5 or Block 18 is a matter of computing the corresponding k-tables as above and then using the k-term output file from Ccorr_k (gas_lwf_l in the example above) as a prompt of prep_spec.

If the spectral range defined in Block 1 is coarsely resolved, two separate spectral files should be provided to SOCRATES: one using a weighting corresponding to a longwave source with the +P flag in the Ccorr_k calls, and the other using a weighting corresponding to a shortwave source with the -s [stellar_spectrum] flag.

Because the -s flag requires to input a stellar spectrum, switching the star requires updating both Block 2 and Block 5.

If the bands are quite narrow, for example with 300 spectral bands, using a single spectral file containing all data blocks would be fine as the k-correlated opacities computed by Ccorr_k would not differ by a large margin without the weighting differentiation. The situations where it becomes important to use different weights for the longwave and shortwave calculations are if the bands are broad, or if the weighting spectrum is ramping one way and the absorption is ramping the other way within a band, such as in the UV with ozone absorption.

5. Running SOCRATES

SOCRATES is run with Cl_run_cdf or CL_run_cdl, the netCDF or CDL versions of the model. Below is an example run which uses input files whose base name is “profile”.

```
Cl_run_cdf -B 'profile' -s {sp_file} -R 1 N_BANDS -ch N_BANDS -I -g 2 -C 5 -u -r -q -o
```

The flag -I indicates that it is solving for a thermal source. For a shortwave spectral file, we would instead use -S to solve for a solar source. Here, the other flags indicate that gas absorption uses a random overlap assumption (-g 2), that clouds are not used (-C 5), that we include continuum absorption (-u) and Rayleigh scattering (-r), and that we use a quadratic variation of the thermal source function over each layer (-q).

6. Outputting the flux contribution function (CFF)

SOCRATES outputs fluxes and temperature tendencies by default. The computation of the contribution function needs to be activated in /src/radiance_core/def_control.F90

Since the source code is changed, the model needs to be recompiled. It will then output the desired .cff output netCDF files on top of the default outputs.

7. Running SOCRATES in Isca

The GCM Isca can be installed using the documentation found [here](#).

Change the paths in the following three files:

src/path_names

src/extra/model/socrates/path_names

src/extra/model/socrates_column/path_names

using [this reference file](#) valid for SOCRATES v. 2403.

Note that each new version of Socrates may require to add new path names. Adapt the paths to each version, e.g. using compilation errors.

Make sure that the external Socrates folder follows the structure socrates_xxxx/trunk/src. Create the trunk if needed.

Add to .bashrc: GFDL_SOC=/path/to/socrates_xxxx/trunk

Remove any trunk/ folder in Isca/src/atmos_param/socrates/src/

After successful compilation, a symlink trunk folder will appear in this location.

Run the test_cases/socrates_aquaplanet.py to check that it works.